

Assumptions, Data Gaps & Recommendations

What was inferred, what is missing, and what to strengthen

Purpose for the AI Assistant

Documents every assumption made while building this knowledge base and the information gaps the AI team should close to raise answer accuracy.

1. Assumptions made

- **Branch code mapping.** Purchase uses short codes (QE, QEN, QCL, QB2) while stock/issuance use full legal names. These were mapped by naming convention; please confirm QB2 vs QBL vs Unit-II mapping with the ERP team.
- **Value windows.** Issuance covers 2025-07-28 to 2026-07-13 and purchases 2025-07-03 to 2026-07-09. Totals are window totals, not all-time; the assistant should state the window when quoting totals.
- **Stock is a snapshot.** The four stock files represent a point-in-time export; there is no timestamp column, so 'current stock' means 'as of the latest export'.
- **'On water' = 'In Transit'.** Mapped the import Current Status value 'In Transit' to the business term 'on water'. Confirm no separate 'sailing' sub-state is expected.
- **Stock Health is numeric.** It is a stock-days coverage ratio, not a Good/Bad label; criticality is taken from AB_Items sheet membership (Critical/Re-Order).
- **Logistics 'Database' sheet is formula-driven.** Static reads return blanks; the record-level source is 'Shipment Master Database' (165 rows). Live figures should be pulled from that sheet or the Dashboard.

2. Data gaps to close

Gap	Impact	Recommended fix
No stock snapshot date	'Current' is ambiguous	Add export timestamp per stock refresh
Item Code missing on some import/logistics rows	Weakens cross-domain joins	Backfill Item Code where possible
Required Date parsing in Purchase	Delay calc depends on it	Standardise date format on export (some rows have ' artifacts)
No unified supplier ID	Same supplier may vary in spelling	Introduce supplier master with canonical IDs
Requisition->Purchase->Import link	Hard to trace demand to fulfilment	Carry Requisition Ref through PO and import file
No planned-arrival in logistics	Delay = actual-ETA only	Store planned arrival separately from ETA

3. Recommendations for the RAG build

- Load **business_dictionary.json** as structured grounding so term->field->logic mappings are deterministic, not left to the LLM.

- Apply the **postgres_schema.sql** and expose the pre-built views (v_inventory_value_by_branch, v_supplier_delay, v_critical_priority, v_on_water) as LangGraph tools – the model calls a view instead of writing raw SQL for common questions.
- Chunk these PDFs by section (H1/H2) with the document code (01–09) as metadata for clean retrieval and citation.
- Enforce the value rules (Available Amoun for stock value; TotalPric for consumption) at the tool layer, not just in prose.
- Add a branch-code resolver as a first-class tool so 'QCL' and 'Qadcast' resolve identically.
- Re-ingest on each data refresh and re-run the profile so the group-scale figures in Document 01 stay current.